

Unit 2

Physical Layer and networking Media

Networking Devices

- Network devices, or networking hardware, are physical devices that are required for communication and interaction between hardware on a computer network.
- These devices are also known as **physical devices, networking hardware, and network equipment.**
- These devices transfer data in a fast, secure and correct way over same or different networks.
- Network devices may be inter-network or intra-network.
- These devices are specific devices, handles digital or electrical connections to perform their exclusive roles very efficiently.

Types of network devices

1. Hub
2. Switch
3. Router
4. Bridge
5. Repeater
6. Access Point

Hub

- The hubs link various networking devices.
Hubs are used to communicate with various network hosts and also for data transferring.
- The transferring of data in a computer network can be done in the form of packets.
- Whenever the data processing can be done from a host to a network hub, then the data can transmit to all the connected ports.
- Similarly, all the ports identify the data path which leads to inefficiencies & wastage.
- Because of this working, a network hub cannot be so safe and secure.
- In addition, copying the data packets on all the ports will make the hub slower which leads to the utilize of the network switch.

Hub: Types

- Network hubs are classified into two types like active hub & passive hub.
- **Active Hub**

These hubs have their own power supply and these hubs are used to clean, increase & transmit the signal using the network.
It works as a wiring center & repeater.
Active hubs play a key role in extending the distance between nodes.
- **Passive Hub**

These hubs collect wiring from the power supply and different nodes of an active hub.
These hubs transmit the signals over the network without improving & cleaning them.
These hubs are not suitable for extending the distance between nodes like an active hub.

Switch

- Similar to a hub, this is also working at the layer in the LAN and a switch is more clever compare with a hub.
- As the hub is used for data transferring, whereas a switch is used for filtering & forwarding the data.
- So this is the more clever technique to deal with the data packets.
- Whenever a data packet is obtained from the interfaces in the switch, then the data packet can be filtered & transmits to the interface of the proposed receiver.

Due to this reason, a switch maintains a content addressable memory table to maintain system configuration as well as memory.

This table is also named as FIB (forwarding information base) otherwise forwarding table.

Router

- A network router is one kind of network device in a computer network and it is used for routing traffic from one network to another.
- A router is a device like a switch that routes data packets based on their IP addresses.
- The router is mainly a Network Layer device.
- Routers normally connect LANs and WANs together and have a dynamically updating routing table based on which they make decisions on routing the data packets.
- Router divide broadcast domains of hosts connected through it.

Router

- The networks could be private to a public company network.
- Consider a scenario,
a router is considered as traffic police at the junction, he directs dissimilar traffic networks to dissimilar directions.

Bridge

- A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of source and destination.

A Bridge in the computer network is used to unite two or more network segments.

- The main function of a bridge in network architecture is to store as well as transmit frames among the various segments.
- Bridges use MAC (Media Access Control) hardware for transferring frames.
- These are also used for connecting two physical local area networks to a larger logical local area network.
used for interconnecting two LANs working on the same protocol.
- It has a single input and single output port, thus making it a 2 port device.

Repeater

- A repeater operates at the physical layer.
- Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted
 - so as to extend the length to which the signal can be transmitted over the same network.
- An important point to be noted about repeaters is that they do not amplify the signal.
- When the signal becomes weak, they copy the signal bit by bit and regenerate it at the original strength.
- It is a 2 port device.

Access Point

- An access point is a device that creates a wireless local area network, or WLAN, usually in an office or large building.
- An access point connects to a wired router, switch, or hub via an Ethernet cable, and projects a Wi-Fi signal to a designated area.
- For example, if you want to enable Wi-Fi access in your company's reception area but don't have a router within range, you can install an access point near the front desk and run an Ethernet cable through the ceiling back to the server room.
- Access points are used for extending the wireless coverage of an existing network and for increasing the number of users that can connect to it.